SIMATS ENGINEERING
ASSESSMENT TOOL 1:
Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Code of Conduct
I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Set 1 - Assessment Tasks
Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

1) Library Book Reservation System

Aim :-

To create a cloud based library book reservation system using SaaS.

Fields :

Book Title, Author, Publication, year, edition, number of copies, Rack number, status (Taken / Returned)

Procedure:

1) Open Zoho Creator and create a new application.
2) Add all the required fields
3) Save and enter book details.
4) View and update book availability.

Result:

The library book reservation system was created successfully.

3) Creating a Virtual machine

Aim:-

To demonstrate virtualization using a type-2 hypervisor.

Software:-

oracle virtualbox.

Procedure:

1) Install oracle virtualbox on the host computer.
2) Download a windows or Linux iso file.
3) Create a new virtual machine.
4) Allocate 1 CPU, 2 GB RAM, and 15 GB storage.
5) Install the operating system and start the VM.

Result:

A virtual machine was created and configured successfully.

2) Payroll Processing System

Aim:-

To create a cloud-based payroll processing system using SaaS.

Fields:

Employee name, Experience in present organization, overal Experience, Basic pay, HRA, DA, TA, Gross pay, Net pay.

Procedure:

1) Create a new application in Zoho Creator.
2) Add the required employee and salary fields
3) Enter employee salary details
4) Calculate gross pay and net pay.

Formula:

⇒ Gross pay = Basic pay + HRA + DA + TA

⇒ Net pay = Gross pay - Deductions.

Result:

The payroll processing system was created successfully.

4) Cloning a virtual machine

Aim:-

To demostrate virtual machine cloning using a Type-2 Hypervisor.

Software:-

oracle virtualbox.

Procedure:

1) open virtualbox and select the existing VM.
2) Click machine → clone.
3) Enter a name for the cloned VM.
4) select full clone and complete the steps.
5) Start the cloned VM and verify that it works

Result:

The virtual machine was cloned and tested successfully.